重新定位 HA 对中的聚合所有权

ONTAP 9

NetApp
February 11, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/disks-aggregates/relocate-aggregate-ownership-task.html on February 11, 2022. Always check docs.netapp.com for the latest.

目录

重新定位 HA 对中的聚合所有权 1
重新定位聚合所有权 1
用于聚合重新定位的命令 3

重新定位 HA 对中的聚合所有权

重新定位聚合所有权

您可以更改 HA 对中节点之间聚合的所有权，而不会中断聚合的服务。

HA 对中的两个节点都以物理方式连接到彼此的磁盘或阵列 LUN 。每个磁盘或阵列 LUN 均归其中一个节点所有。尽管在发生接管时磁盘所有权会暂时发生变化，但聚合重新定位操作可能是永久的（例如，为负载平衡而执行的操作），也可能是临时的（例如，在接管过程中执行的操作） 将聚合中所有磁盘或阵列 LUN 的所有权从一个节点更改为另一个节点。所有权更改无需执行任何数据复制过程，也无需物理移动磁盘或阵列 LUN 。

关于此任务

- 由于在聚合重新定位操作期间会以编程方式验证卷计数限制，因此无需手动检查此限制。

  如果卷数超过支持的限制，则聚合重新定位操作将失败，并显示相关的错误消息。

- 在源节点或目标节点上正在执行系统级操作时，不应启动聚合重新定位；同样，也不应在聚合重新定位期间启动这些操作。

  这些操作可以包括以下内容：

  - 接管
  - 交还
  - 关闭
  - 另一个聚合重新定位操作
  - 磁盘所有权更改
  - 聚合或卷配置操作
  - 存储控制器更换
  - ONTAP 升级
  - ONTAP 还原
- 如果您使用的是 MetroCluster 配置，则不应在执行灾难恢复操作（ *switchover* ， *healing* _ 或 _*switchback* ）时启动聚合重新定位。
- 如果您拥有 MetroCluster 配置并在已切换的聚合上启动聚合重新定位，则此操作可能会失败，因为它超过 DR 配对节点的卷限制计数。
- 您不应对已损坏或正在进行维护的聚合启动聚合重新定位。
- 在启动聚合重新定位之前，应保存源节点和目标节点上的所有核心转储。

步骤

1. 查看节点上的聚合以确认要移动的聚合，并确保它们处于联机状态良好：

   `storage aggregate show -node` *`source-nod`*

   以下命令显示了集群中四个节点上的六个聚合。所有聚合均已联机。Node1 和 Node3 形成一个 HA 对，Node2 和 Node4 形成一个 HA 对。

```
cluster::> storage aggregate show
Aggregate     Size Available Used% State   #Vols  Nodes  RAID Status
--------- -------- --------- ----- ------- ------ ------ -----------
aggr_0     239.0GB   11.13GB   95% online       1 node1  raid_dp,
                                                         normal
aggr_1     239.0GB   11.13GB   95% online       1 node1  raid_dp,
                                                         normal
aggr_2     239.0GB   11.13GB   95% online       1 node2  raid_dp,
                                                         normal
aggr_3     239.0GB   11.13GB   95% online       1 node2  raid_dp,
                                                         normal
aggr_4     239.0GB   238.9GB    0% online       5 node3  raid_dp,
                                                         normal
aggr_5     239.0GB   239.0GB    0% online       4 node4  raid_dp,
                                                         normal
6 entries were displayed.
```

2. 使用问题描述命令启动聚合重新定位：

   s存储聚合重新定位 `start -aggregate-list` *`aggregate-1`* ， *`aggregate-2`* … `-node source-node -destination` *`destination-nod`*

   以下命令将聚合 aggr_1 和 aggr_2 从 Node1 移动到 Node3 。Node3 是 Node1 的 HA 配对节点。聚合只能在 HA 对中移动。

```
cluster::> storage aggregate relocation start -aggregate-list aggr_1,
aggr_2 -node node1 -destination node3
Run the storage aggregate relocation show command to check relocation
status.
node1::storage aggregate>
```

3. 使用 `storage aggregate relocation show` 命令监控聚合重新定位的进度：

   `storage aggregate relocation show -node` *`source-nod`*

   以下命令显示要移动到 node3 的聚合的进度：

```
cluster::> storage aggregate relocation show -node node1
Source Aggregate   Destination    Relocation Status
------ ----------- -------------  ------------------------
node1
       aggr_1      node3          In progress, module: wafl
       aggr_2      node3          Not attempted yet
2 entries were displayed.
node1::storage aggregate>
```

重新定位完成后，此命令的输出将显示每个聚合的重新定位状态为 `Done` 。

用于聚合重新定位的命令

有特定的 ONTAP 命令可用于在 HA 对中重新定位聚合所有权。

如果您要 ...	使用此命令 ...
启动聚合重新定位过程	s存储聚合重新定位开始
监控聚合重新定位过程	s存储聚合重新定位显示

"ONTAP 9 命令"

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.